

Compressor

One compressor, four voicings: punchy VCA, smooth bus glue, transparent ride and optical.



Overview

Compressor is a single compressor with four characters, chosen with the VOICING buttons. VCA is a punchy compressor that measures loudness (RMS), with a ratio that goes past infinity into negative ratios and a touch of even harmonics under heavy reduction. BUS is a stereo-linked glue compressor that reads the peaks and stays smooth and clean. RIDE is a transparent 'easy ride' leveller with a wide knee and timing that adapts to the music. OPTO behaves like an optical cell: a soft attack and a two-stage release that lets go quickly at first, then slowly.

With AUTO on, attack and release follow the programme, and the release lengthens when the compressor keeps working, so sustained passages don't pump. The SIDE-CHAIN section either stops the lows from driving the compressor (SC HPF) or compresses only the lows (SPLIT), and MIX blends in the dry signal for parallel compression. The transfer curve shows exactly what the compressor is doing to the level as you play.

Reach for it for

- Vocals that need levelling (RIDE, OPTO)
- Drums and drum buses (VCA)
- Gluing a mix or group bus (BUS)

- Tightening only the low end of a bass or mix (SPLIT)
- Parallel crush and pumping effects with negative ratios (VCA, MIX)

Controls

The graph section

SIGNAL	Lights green when audio is arriving.
INPUT	The input level (average) in dB.
Transfer curve	The input-to-output curve from -60 to 0 dB, including the knee. The small black triangle under it marks the THRESHOLD. The red square rides on the curve with the live level, and the red arrow at the right shows the output level after MAKEUP.
Gain reduction meter	24 lights lit from the right, showing how many dB the compressor is taking off (0 to -24).

COMPRESSOR

RATIO	How strongly the compressor works once the level is over the threshold. 1:1 does nothing; 2:1 and 4:1 are moderate; INF:1 is limiting. In the VCA voicing the ratio goes on past INF into negative ratios (down to -1:1): the louder the input, the quieter the output, for pumping and backwards effects. The other voicings stop at INF:1.
KNEE	The soft knee, 0 to 18 dB (6 dB by default). 0 is a hard knee; larger values ease the ratio in gradually around the threshold. RIDE always uses at least 10 dB and OPTO at least 12 dB.
MAKEUP	Make-up gain, 0 to +24 dB, to bring the level back up after compression.
THRESHOLD	The level where the compressor starts working, -50 to 0 dB (-18 dB by default). Turn it down for more compression.

MIX

Parallel compression: blends the dry signal back in. 100% is fully compressed; lower settings keep the dry transients under the compressed sound.

Timing and voicing

ATTACK

How fast the compressor reacts, 0.1 to 100 ms (10 ms by default). The display reads AUTO when AUTO is on, and OPTO in the OPTO voicing, where the knob has no effect.

RELEASE

How fast it recovers, 20 to 2000 ms (150 ms by default). Reads AUTO or OPTO in the same way.

AUTO

On by default. Attack and release follow the programme instead of the knobs: the release gets longer the more the compressor keeps working, so sustained passages don't pump. Always lit in the OPTO voicing, which has its own fixed timing.

VCA / BUS / RIDE / OPTO

The voicing. VCA: punchy, measures loudness (RMS), allows negative ratios and adds a touch of even harmonics under heavy reduction. BUS: stereo-linked, reads the peaks, smooth and low colour. RIDE: transparent leveller with a wide knee and adaptive timing. OPTO: optical cell with a soft attack and a two-stage release (quick, then slow).

LINK

On by default: both sides get the same gain, so the stereo image stays still. Off: each side is compressed on its own. Always lit in the BUS voicing, which is always linked.

SIDE-CHAIN

SC HPF

The compressor ignores the lows below the frequency when deciding how much to compress (a 12 dB per octave high-pass on what it listens to). The whole signal is still compressed. Use it so the bass doesn't make everything pump. This is the default.

SPLIT

Only the lows below the frequency are compressed; everything above passes untouched. The split is a 24 dB

per octave crossover that sums back flat when nothing is being compressed. Use it to tighten the low end of a bass or a mix on its own.

Frequency

The side-chain frequency, from OFF (fully left, 20 Hz) to 2 kHz (80 Hz by default). The labels beside it remind you what the mode does: DETECTOR HIGH-PASS or LOWS ONLY COMPRESSED.

IN / OUT

TRIM

Input level before anything else, from -24 to +12 dB.

GUARD

Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right and is lit while it is on. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light glows brighter while it is catching peaks.

SAT

Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

OFF / CLIP / LIMIT

The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.

Drift

A slow random wander: the threshold and the timing breathe slightly, like an analogue unit.

DRIFT

How much drift, 0 to 100%. 50% (the default) is the standard amount, 100% is twice that, 0 switches it off. More drift also makes the wander a little faster.

Output

Meter	Output level, left and right.
LEVEL	Output level, -24 to +12 dB.

The preset row and colour strip

PRESETS	Opens the list of factory presets (a click on the display does the same). Choosing one sets the compressor for that job; drift, output level and the IN / OUT section are left as they are. The last entry resets everything to the defaults.
◀ / ▶	Step to the previous or next preset.
Colour swatch	Changes the faceplate colour: GREEN, BLUE or one of the GFX themes.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW / FLAT	Switches between the hardware-style panel and the flat look.

Presets

Vocal Ride	RIDE at 3:1 with a wide knee and AUTO timing, for even, natural vocals.
Drum Smash	VCA at about 6:1, fast 3 ms attack and 80 ms release, a tight 2 dB knee: punchy, aggressive drums.
Bass Glue	VCA at 4:1 with AUTO timing; SPLIT at 120 Hz so only the lows are compressed.
Bus Glue	BUS at 2:1 with a 30 ms attack and 300 ms release: classic gentle bus glue.

Negative Pump	VCA past INF at -2:1 for rhythmic pumping.
Gentle Level	RIDE at 2:1 with a very wide 15 dB knee and AUTO timing: barely-there levelling.
Parallel Crush	VCA crushing hard (about 10:1, 1 ms attack, hard knee), blended in at 40%.
Mastering Touch	BUS at a gentle 1.5:1 with AUTO timing, just 1 dB of make-up.
Opto Vocal	OPTO at 4:1: soft, smooth vocal levelling.
Opto Lows Only	OPTO at 5:1 with SPLIT at 160 Hz: smooth control of the low end only.
Reset all to defaults	Every control back to its starting value.

Tips

- Watch the transfer curve: if the red square sits far up the flat part of the curve, you are compressing more than you probably need.
- If the low end makes the whole mix duck, use SC HPF at 80-120 Hz.
- SPLIT is a quick way to steady a boomy bass or kick without dulling the top.
- Negative ratios sound best on sustained material and with MIX below 100%.
- Try the same settings in each voicing: VCA for punch, BUS for glue, RIDE and OPTO for smooth, transparent levelling.

File: gfx-compressor.jsfx